

1 Three Dimensional Coordinate Systems and Vectors

1.5 Equations of Lines and Planes

1.5.1 Lines

Parametric equations of a line, where (x_0, y_0, z_0) is a point on the line, and $\langle a, b, c \rangle$ is a vector parallel to the line.

$$x = x_0 + at \quad y = y_0 + bt \quad z = z_0 + ct.$$

Example 1.5.1 Find a vector equation and the parametric equations of the line passing through $(1, 3, 2)$ and parallel to $\langle 1, 4, -2 \rangle$.

Symmetric equations of a line where (x_0, y_0, z_0) is a point on the line, and $\langle a, b, c \rangle$ is a vector parallel to the line.

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Example 1.5.2 Find the symmetric equations of the line whose vector equation is given by

$$\vec{r}(t) = (3, 2, 1) + t \langle -1, 0, 3 \rangle.$$

Example 1.5.3 Show that the lines L_1 and L_2 whose equations are

$$L_1 : x = 1 + t, \quad y = -2 + 3t, \quad z = 4 - t$$

$$L_2 : x = 2s, \quad y = 3 + s, \quad z = -3 + 4s$$

Are skew lines (that is, they are not parallel, and do not intersect).

Example 1.5.4 Line segments.

1.5.2 Planes

Example 1.5.5 Find the equation of the plane which passes through the point $(2, 4, -1)$ with normal vector $\vec{n} = \langle 2, 3, 4 \rangle$.

Example 1.5.6 Find the equation of the plane which passes through the points $P(1, 3, 2)$, $Q(3, -1, 6)$ and $R(5, 2, 0)$.

Example 1.5.7 Find the point at which the line $x = 2 + 3t$, $y = -4t$, $z = 5 + t$ intercepts the plane $4x + 5y - 2z = 18$.

Example 1.5.8 Find the angle between the two planes $x + y + z = 1$ and $x - 2y + 3z = 1$.

1.5.3 Distances

Example 1.5.9 Find the distance from the point $(4, 0, 1)$ to the plane $x - y + z = 1$.

Example 1.5.10 Find the distance between the two planes $10x + 2y - 2z = 5$ and $5x + y - z = 1$.